ChipLayoutBench

A Multimodal Benchmark for Vision-Language Models in Semiconductor Layout Understanding

Serim Ryou [†] Jayeon Yoo [†] Minjae Jeong [◆]
Won Hwa Kim [◆] Seong-Jin Park [†]

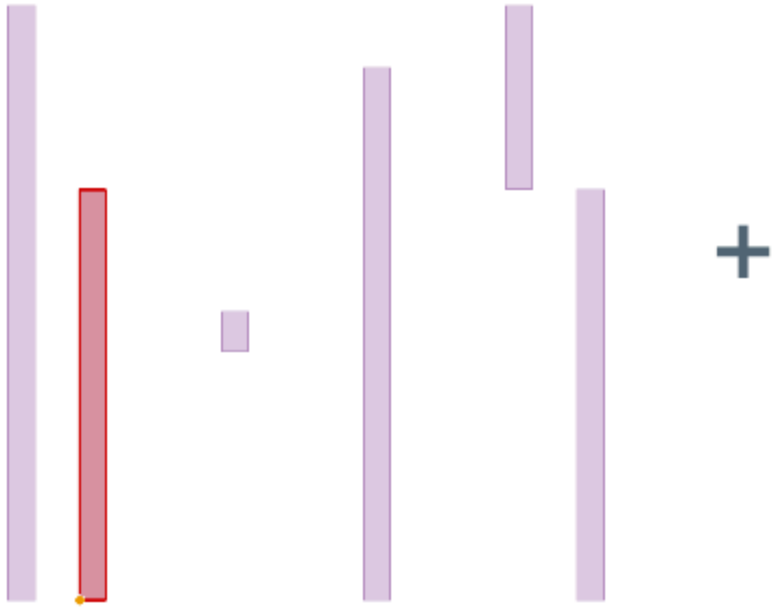
[†] AX/PI Center, Samsung Electronics
[◆] POSTECH

slryou41.github.io/ChipLayoutBench/

Wrong polygon match, wrong coordinate

Input: layout image + GDS-derived text

Question: What is the x-coordinate of the highlighted vertex?



+

poly_00001

LAYER 13 DATATYPE 0
x = 320.78

Wrong record

Model: 320.78

poly_00002

LAYER 13 DATATYPE 0
x = 320.97

Actual target

Answer: 320.97

Appendix Figure 29: actual model response

Three representations of the same layout

TEXT

GDS geometry as text

POLYGON poly_00002
LAYER 13 DATATYPE 0
(320.97, 255.485) ...

IDs, layers, exact vertices

IMAGE + METADATA

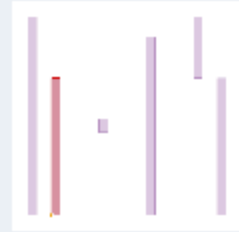


Layout rendering

Reference coordinate
Scale: 1 nm per pixel
Layer-to-color mapping

BOTH

GDS text +



+ spatial metadata

Match target to its GDS record

What do we ask the models?

Task category	Example question	Answer
Count & describe	How many polygons have 4 vertices?	**4**
Find & retrieve	Which polygon is closest to the top-left corner?	**poly_00042**
Measure & reason	What is the extruded area of poly_00038?	**1.5314**
Check layout quality	How many m1 width violations at 0.018 µm?	**3**
Match image to text	Which ID matches the highlighted polygon?	**poly_00082**

2,115 layouts **6,808 test questions**

Answers computed from geometry; human-checked shape labels

Table 1 and Sections 3.1–3.5; examples use different samples

Adding images lowers scores for 7 of 8 models



Table 2: average score as reported in the paper

Even image-helpful questions show a decline

Question subset	Text	Both
Text-sufficient	**82.4%**	**81.6%**
Image-helpful	**56.2%**	**52.3%**

Images may provide useful cues. Answers remain derivable from text.

Table 3 and Section 4.2; macro-average exact match across eight models

What the benchmark reveals

Observed result

Visual context does not reliably improve exact layout reasoning.

What the failure case illustrates

A plausible coordinate can come from the wrong polygon.
The visual target must link to the correct GDS record.

Current coverage: flattened geometry from a limited set of circuits and processes